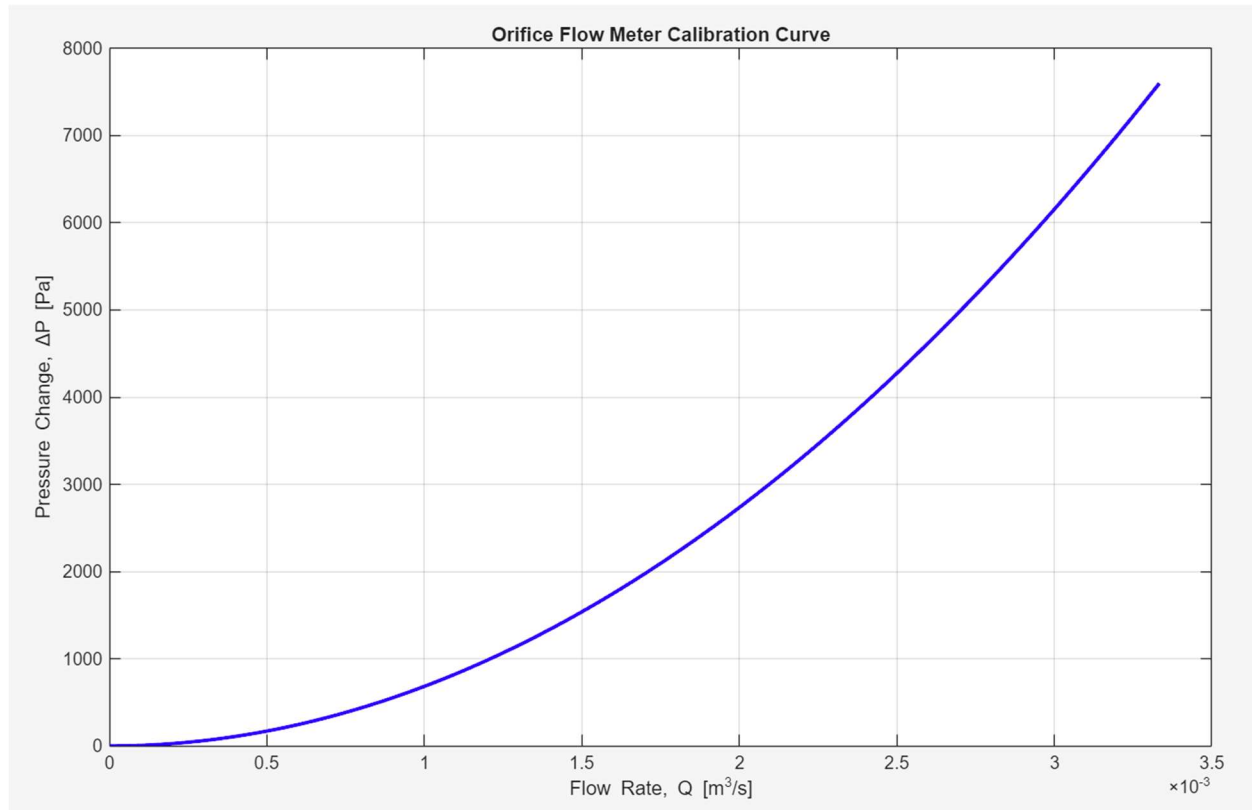


Q1:



As seen in the following graph, there is a non-linear relationship between flow rate and pressure change. As a result, the slope is not constant. If $\Delta P \propto Q^2$, then $\frac{dP}{dQ} \propto \frac{d}{dQ}(Q^2) = 2Q \neq \text{const.}$

As calculated in the MATLAB script:

```
g = 9.81;
Qmax = 200/60000; %m^3/s
dh = 1/(2*g).*(Qmax/(C*A)).^2;
disp(['Max. Head = ', num2str(dh), ' [m]'])

dPmin = 98.1/2; %Pascals
Q = C*A*sqrt(2*dPmin/1000);
disp(['Min. Flow Rate = ', num2str(Q), ' [m^3/s]'])

>> Day4OrificeMeter
Max. Head = 0.77456 [m]
Min. Flow Rate = 0.00026782 [m^3/s]
```

Maximum head at highest flow rate is 0.77456 m, and minimum flow rate reasonably measured with a resolution of 10 mmH₂O is 0.00026782 m³/s.

Q2: *AI was used for the following report. As requested, below is the exact prompt and resulting report.*

The following report for Student 3 needs to be graded based on the following rubric. the rubric contains what should be in the report, along with noting general grammar and convention. First, give the student a grade based on how well they performed the analysis. Then, give feedback on what areas need improvement. Finally, generate a fully edited report. First rule: do not remove any information, keep the core work as it is. Second rule: complete this as if you were grading and editing as a professional engineer (i.e. use active voice, be technical and straight to the point, minimal fluff/narrative, etc).

Francis Turbine Lab Report — Student 3 (Edited)

Abstract

The Francis turbine converts the kinetic and pressure energy of water into mechanical energy. Water enters the turbine radially, imparting lift and impulse forces on the runner blades that rotate the shaft. Francis turbines are commonly used in dams to generate electricity for surrounding areas. Turbine performance is defined primarily by power output and efficiency — the ratio of mechanical power produced to hydraulic power supplied.

This study independently varied three parameters — guide vane angle, applied torque, and water flow rate — to quantify their effects on power output and efficiency. For each parameter setting, five quantities were recorded: change in volume, change in time, change in pressure, applied load, and output shaft speed. Volume and time measurements were repeated three times per setting to reduce random error. These measurements were used to calculate work input and work output, from which efficiency was determined at each test point.

Data were collected using a Francis turbine test rig provided by Northern Arizona University. Testing proceeded in three stages: guide vane angle was varied across five settings to identify the angle of maximum efficiency; resistive torque was then varied across six settings at the optimal guide vane angle; and flow rate was varied across three settings, also at the optimal guide vane angle.

Results show that maximum efficiency (16%) occurred at a guide vane angle of 30°. Efficiency was more sensitive to resistive torque and flow rate than to angular velocity and power output: efficiency decreased as torque decreased and angular velocity increased, and efficiency decreased as flow rate decreased and power output increased. These trends agree with the theoretical performance equations. The maximum efficiency measured across all tests was 45%. Instrument uncertainty — primarily reflective interference in tachometer RPM measurements — was the largest source of error in the analysis.

Conclusion

This lab characterized Francis turbine efficiency across three independently varied parameters: guide vane angle, resistive torque on the output shaft, and water flow rate. Efficiency is a critical design parameter for real-world hydroelectric applications. The lab objective was to quantify these parameter-efficiency relationships through a series of controlled tests. Five measurements were recorded at each parameter setting to calculate work input, work output, and resulting efficiency.

Figure 6 presents the guide vane angle results and shows a maximum efficiency of 16% at a 30° vane angle. This result suggests a weak dependence of efficiency on vane angle; however, the Prony brake was loaded with a mass difference of 0.175 kg rather than the recommended 0.6 kg, because the higher mass produced enough friction to stall the shaft. A larger mass difference would likely reveal a stronger relationship between vane angle and efficiency.

Figure 7 shows that efficiency depends more strongly on resistive torque than on angular velocity. Equation 10 indicates that efficiency is directly related to both variables, and in theory a proportional decrease in torque and increase in velocity should hold efficiency constant. The data instead show efficiency decreasing at approximately the same rate as torque, indicating that torque change dominates angular velocity change in its effect on efficiency. Maximum efficiency in this test was 45%.

Figure 8 shows that efficiency depends more strongly on flow rate than on power output. Because efficiency is a function of power output, and power output is itself a function of torque and angular velocity, maximizing efficiency requires low resistive torque and high power output — competing conditions. Maximum efficiency therefore occurs at a balance between the two.

Error in this analysis was primarily instrumentation-driven. Tachometer vibration from shaft rotation scattered the RPM data and reduced measurement accuracy. The bucket-and-timer flow method also introduced error due to the 1 L resolution of the volume bucket, which limited measurement precision. Future testing should use a steadier tachometer and a finer-resolution volume measurement to reduce error. Because turbine efficiency directly affects the economic viability of hydroelectric power generation, minimizing this error is essential for accurate performance characterization.

Q3:

Table __: RTD sensor output vs. standard reference temperature from -80°C to 190°C, with error bars showing u_{std} .

<i>Temp. Standard [C]</i>	<i>u_{std} [C]</i>	<i>RTD Sensor [C]</i>
- 79.998	0.024	-80.01
- 0.001	0.01	0
50	0.01	50.01
189.999	0.04	189.97